Ruhan Bhavsar

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Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science - Mechatronics Engineering — *Sept 2024 - Present*

Experience

Engineering Intern — Jan 2026 - Apr 2026
MPI Morheat — *Toronto, ON*

- Developed a **React-based** configurator for flange, screwplug, and PTFE heater drawings, reducing custom drawing generation time by **95%** from approximately **20 minutes to 1 minute** for customer proposals.
- Developed a low-cost (<**$500**) automated **three-axis label cutter** using **stepper motors**, **3D-printed components**, and an **STM32 microcontroller** programmed in **C++**, **saving 5–10 hours of manual labor per week**.

Engineering Intern — May 2025 - Aug 2025
Vexos — *Markham, ON*

- Built manufacturing tools with **Python**, **SolidWorks**, and **3D printing** that reduced equipment downtime by **40 hours monthly**.
- Automated calibration workflow in **Asset Essentials**, reducing overdue work orders by **30%** and ensuring **100% compliance** with industry standards.
- Created standardized instructions that cut operator training time by **2 hours** and reduced assembly-line errors by **25**%.

Firmware Developer — Sept 2024 - Present
Midnight Sun Solar Car Team — *Waterloo, ON*

- Maintained embedded software for **8 microcontrollers**, managing critical **real-time tasks** and **ADC sampling** with a **10 ms latency target** to ensure system-wide synchronization.
- Implemented a **Hardware-in-the-Loop (HIL)** testing framework using **Pytest** and **Python scripts**, reducing manual testing time by **80**%.
- Optimized firmware builds with **SCons** and **Vagrant**, reducing deployment time by **10 minutes**.

Projects

Natural Disaster Severity Mapping | *React, Flask, MongoDB, PyTorch, ResNet34*

- Built a full-stack web app using **React**, **Flask**, and **MongoDB** to assess earthquake severity from user-uploaded images.
- Trained a **ResNet34 CNN model** in **PyTorch** to classify and rank natural disaster severity with an accuracy of **95**% on a dataset of **8,500 images**.
- Integrated **Mapbox API** for dynamic geolocation visualization and secure user authentication for submissions.
- Deployed a **responsive web interface** enabling **real-time map updates** and interactive data display.

Robotic Hand | *Python, MediaPipe, ESP32*

- Designed and 3D printed a robotic hand with six servo-driven fingers for motion replication.
- Developed a **Python-based computer vision system** using **MediaPipe** to detect finger states with **45ms end-to-end latency**, enabling near-instant replication of human hand movements.
- Implemented **real-time serial communication** with **PySerial** to transmit detected finger coordinates to the **ESP32**, translating virtual states into precise servo movements.

Rescue Robot | *C++, RobotC, AutoCAD and SolidWorks*

- Programmed a Lego EV3 robot to autonomously navigate a grid using a **Depth-First Search (DFS)** algorithm for optimal pathfinding.
- Designed and simulated robot chassis and components in **SolidWorks** and **AutoCAD**.

Technical Skills

Programming Languages: Python, JavaScript, TypeScript, C++, C, Kotlin, SQL, HTML/CSS, MATLAB, VHDL
Frameworks/Libraries: React, Flask, Node.js, Jetpack Compose, TailwindCSS, PyTorch, FastAI, MongoDB, PySerial
Tools: Android Studio, Git, Linux, Docker, Vagrant, SCons, SolidWorks PDM, Bambu Studio
Concepts: REST APIs, Full-Stack Development, Embedded Systems, Unit Testing, CI/CD, Version Control (Gitflow)